

Validation Note: Sona Urea Forecast

Company-Calibrated Synthetic Dataset for Academic Analysis — illustrative figures, calibrated to Fauji Fertilizer Company Limited's disclosed scale, not actual company records.

Headline result. On 12 weeks it never saw, the selected averaged forecast missed by 14.7% of volume (WAPE), with a bias of +2.1 KT a week and a tracking signal inside ± 4 in every week; no holdout demand value was used to fit any model.

Holdout design

Item	Setting
Dataset	Frozen synthetic dataset v2.0: 156 weekly rows, 2 Jan 2023 to 22 Dec 2025, checksum 0d23533...
Training period	144 weeks, 2 Jan 2023 to 29 Sep 2025
Holdout period	12 weeks, 6 Oct to 22 Dec 2025 (the final block, as the brief requires)
Forecast method	All 12 holdout weeks forecast once from 29 Sep 2025 (fixed origin); no recursive updating with actual values
Models compared	Naive, seasonal naive, 4-week moving average, SES (α 0.3 and 0.6), seasonal decomposition + Event_Flag, and the average of decomposition and seasonal naive
Selected model	Average of seasonal decomposition and seasonal naive (equal weights, not tuned)
Error convention	Error = Actual – Forecast; positive bias means under-forecasting
Robustness test	The same comparison repeated over 17 rolling 12-week windows (origins every 4 weeks from Jul 2024)

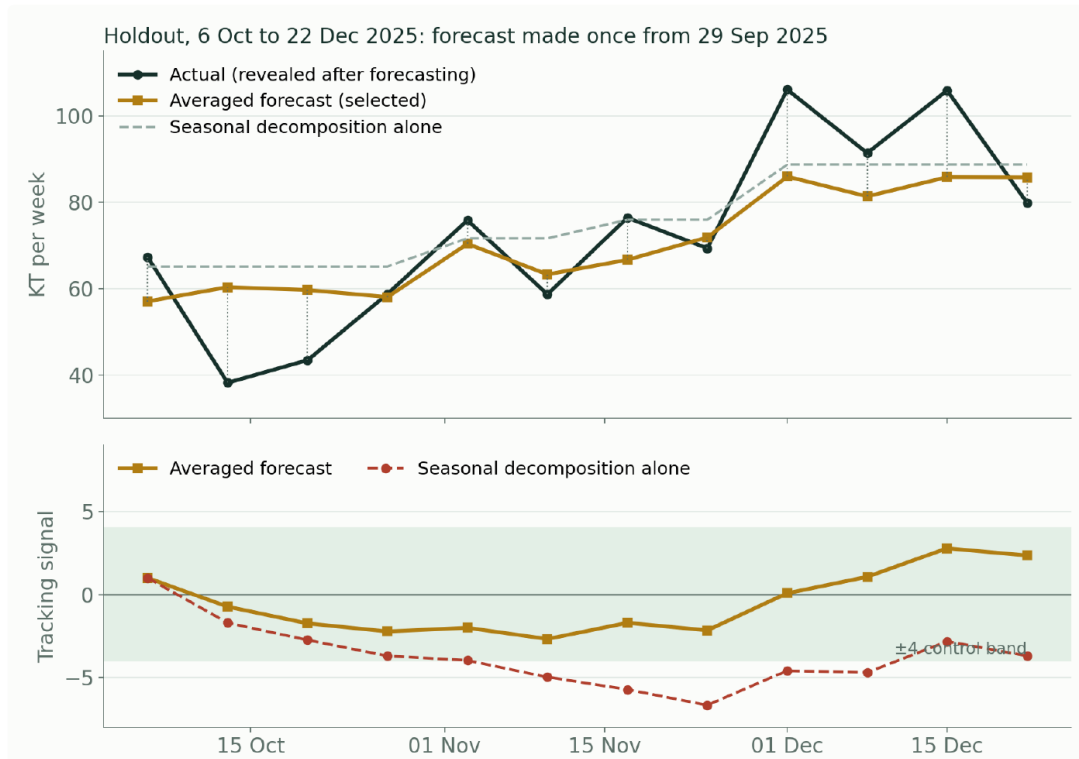
Leakage checks

All checks pass.

Check	Result
Date order	156 consecutive Mondays, every gap 7 days, no duplicates or missing values
Train/test separation	Training ends 29 Sep 2025; holdout starts 6 Oct 2025; no week in both
Seasonal naive inputs	Holdout forecasts read 7 Oct to 23 Dec 2024, all inside the training data
Scramble test	Replacing holdout demand with random numbers changes the forecasts by exactly 0
Delete test	Removing the holdout demand column leaves the forecasts unchanged
Calendar-only test	Forecasts rebuild from Week_Start_Date, Event_Flag and Merger_Flag alone, all known in advance
Plausibility	Holdout WAPE (14.7%) is close to the rolling-window average (18.1%), not suspiciously low; errors run in both directions

Five largest errors and diagnosis

Rival discounting and Rabi-peak scheme weeks explain most of the miss; none is a stockout or data problem.



Holdout, 6 Oct to 22 Dec 2025: forecast made once from 29 Sep 2025. Top: actual vs forecasts (KT per week). Bottom: tracking signal against the ± 4 control band.

#	Week	Actual (t)	Forecast (t)	Error (t)	Share of error	Diagnosis
1	13 Oct 2025	38,180	60,308	-22,128	17.3%	Genuinely unpredictable: rival dealer discount campaign
2	1 Dec 2025	106,060	85,917	+20,143	15.8%	Promotion or peak level: 4.2% scheme at the Rabi peak; one season cannot separate the two
3	15 Dec 2025	105,860	85,792	+20,068	15.7%	Promotion or peak level: 7.7% scheme at the Rabi peak
4	20 Oct 2025	43,390	59,648	-16,258	12.7%	Genuinely unpredictable: second week of the rival campaign; demand recovered the next week
5	6 Oct 2025	67,270	57,023	+10,247	8.0%	Promotion effect: 5.4% scheme week; the forecast has no promotion input (model weakness)

Together these five account for 69.5% of holdout error. There were no stockout weeks in the holdout. The fixable weakness is the missing promotion input; the rival campaign calls for contingency stock, not a better formula.

Robustness and disclosures

- **Rolling test:** across 17 rolling windows the averaged forecast had the lowest mean WAPE (18.1%) and bias; seasonal decomposition alone won only 9 of 17 windows against seasonal naive and reached 43–49% WAPE at the 2025 turning point.

- **Tracking-signal path:** seasonal decomposition alone ended the holdout at -3.7 but breached -4 in weeks 5 to 10; the averaged forecast stayed between -2.7 and $+2.8$.
- **Indirect leak:** the merger scope factor ($\times 1.184$) uses full-year 2025 annual report figures that include the holdout quarter. A training-only factor ($\times 1.09$) changes decomposition's holdout WAPE only from 14.6% to 14.8%.
- **Holdout reuse:** the holdout was also used to compare model variants, so the rolling-window result is the primary evidence for the model choice.
- **Synthetic data:** all demand values are synthetic, so this validation tests the method, not FFC's actual market.